

Zelin Li

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Education

- University of Minnesota, Twin Cities**, Bachelor's Degree in Computer Science Sept 2025 – May 2027
• **GPA: 4.0/4.0**
- Beijing University of Posts and Telecommunications (BUPT)**, Artificial Intelligence Sept 2023 – June 2025
• GPA: 3.85/4.0, **Rank: 1/180**

Research Interests

LLM Inference and Post-Training · Zeroth-Order Optimization · Algorithm-System Co-Design · Optimization Theory

Publications & Preprints

Inference-Native Zeroth-Order Optimization.

Zelin Li, Caiwen Ding.
[arXiv:2605.28760](https://arxiv.org/abs/2605.28760) [↗](#), 2026.

Sample-Stable Subspaces and Mini-Batch Noise in Zeroth-Order Fine-Tuning.

Zelin Li, Caiwen Ding.
Preprint, 2026.

DynamicMem: A Long-Horizon Memory Benchmark in Real-World Settings.

Wenya Xie, Shengming Zhou, **Zelin Li**, et al.
[arXiv:2606.22877](https://arxiv.org/abs/2606.22877) [↗](#), 2026. Under review at NeurIPS 2026.

Research Experience

Research Assistant, University of Minnesota, Twin Cities Apr 2026 – Present
Advisor: Prof. Caiwen Ding

Inference-Native ZO Systems

- Formulated ZO as candidate-state queries and developed an inference-native execution path that separates algorithmic probe semantics from runtime realization.
- Built the vLLM-based prototype with LoRA-side candidate/update state, lazy and banked updates, packed variable-length scoring, and mixed foreground inference/ZO execution.

ZO Stability and Mini-Batch Noise

- Studied why structured, low-dimensional ZO remains batch-sensitive even after the search geometry becomes highly stable across samples.
- Designed large-batch, fixed-subspace, and matched-direction experiments on OPT-13B and Qwen3 to isolate mini-batch objective noise from search-space variation.

Research Assistant, University of Minnesota, Twin Cities Nov 2025 – Mar 2026
Advisor: Prof. Zirui Liu; worked with Wenya Xie

Long-Horizon LLM Memory Benchmark

- Implemented and evaluated HippoRAG and SimpleMem; contributed to benchmark-data construction and extensive manual auditing of generated data.
- Used audit findings to support data and algorithm revisions, and contributed to experimental analysis and manuscript preparation.

Honors & Awards

National Scholarship, Ministry of Education of China (5/470) 2024
Silver Medal, International Collegiate Programming Contest (ICPC), Asia Shenyang Regional (84/399) 2024
Bronze Medal, China Collegiate Programming Contest (CCPC), Harbin 2024
Bronze Medal, Beijing Collegiate Programming Contest 2024
First Prize, National Olympiad in Informatics in Provinces (NOIP), Shanxi 2021

Technical Skills

Programming: Python, C++ **Frameworks:** PyTorch, vLLM, Hugging Face Transformers **Tools:** Linux, Git